

$$Z(\Theta) = \varepsilon \colon A/[A,A] \rightarrow \mathbf{k}$$

$$Z\left(\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}\right) \colon A_1 A_{A_2} \otimes A_3^{\text{op}} A_{A_4^{\text{op}}} \rightarrow A_1 A_{A_3} \otimes A_4 A_{A_2} \colon a \otimes b \mapsto \sum ax_i \otimes bx^i$$

$$Z\left(\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}\right) \colon A_1 A_{A_2} \otimes A_3^{\text{op}} A_{A_4^{\text{op}}} \rightarrow A_1 A_{A_3} \otimes A_4 A_{A_2} \colon a \otimes b \mapsto \sum ax_i \otimes bx^i$$

$$Z(\Theta)=[w^{-1}]\in A/[A,A]$$